Bioenergetics and electron carriers

The ability to obtain, store and use energy is characteristic of living organisms. **Bioenergetics** is the study of how organisms obtain and use energy.

The energy transformations that take place in living cells obey the same laws of thermodynamics that govern all natural processes. The basic principles of thermodynamics are:

1. First law of thermodynamics or law of conservation of energy: in any physical or chemical change the total energy in an isolated system remains constant.
2. Second law of thermodynamics: any physical or chemical change proceeds spontaneously in such a way that the entropy increases.

Living organisms maintain their internal organization by exchanging matter and energy with their surroundings. Although entropy may decrease locally within an organism, the total entropy of the organism and its environment increases.



Watch this video as an introduction to the relationship between life and energy

Spontaneity of biological processes: Gibbs free energy

A fundamental aspect in the study of metabolic processes is the ability to determine in which direction they will take place spontaneously.

Suppose a reaction in which compounds A and B give rise to compounds C and D:

$$A + B \leftrightarrows C + D$$

It is important to determine whether the process will progress towards the formation of C and D, or towards the formation of A and B; as well as whether a change in the physiological conditions of the cell, such as an increase in compound C, can change the direction of the process.

The second law of thermodynamics determines that processes take place spontaneously in the direction in which the entropy increases. However, entropy changes are not easily measurable and it is required to know both the entropy change of the reaction and the environment.

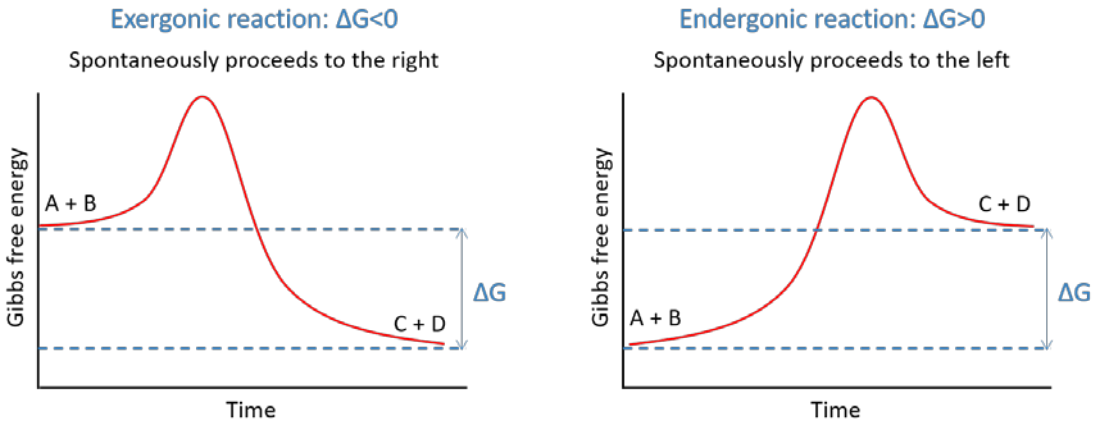
In the case of biological processes, the **Gibbs free energy change** (**ΔG**) is the parameter used to determine in which direction a reaction will proceed.

$$\Delta G = \sum G_{productos} - \sum G_{reactivos}$$

When a reaction releases free energy (that is, ΔG has a negative sign) the reaction is said to be **exergonic**. When a reaction requires free energy (ΔG has a positive sign) the reaction is said to be **endergonic**.

At constant temperature and pressure (which applies to biological processes), **a reaction proceeds spontaneously in the direction in which ΔG<0**.

- If the reaction *A+B⇆C+D has a ΔG<0,* the exergonic process (thermodynamically favorable) will be the formation of C and D, so the reaction will proceed spontaneously to the right.
- *If the reaction A+B⇆C+D has a ΔG>0,* the exergonic process (thermodynamically favorable) will be the formation of A and B, so the reaction will proceed spontaneously to the left.
- *If the reaction A+B⇆C+D has a ΔG=0,* the process will not go to the right or to the left, but will be in equilibrium.



A process cannot happen spontaneously if ΔG>0 (endergonic reaction). **Organisms maintain their internal organization by obtaining free energy from nutrients or sunlight and ultimately releasing energy to the environment, mainly as heat**. By coupling endergonic processes to sufficiently exergonic reactions, cells obtain an overall process with a negative ΔG.

Determination of the free energy change of a reaction

Free energy values of compounds are not easy to measure; however, free energy changes of processes are. The free energy changes of a multitude of processes, including the main metabolic reactions, have been determined experimentally. These values can be consulted on biochemical databases and reference tables.

Since the experimental conditions affect ΔG, by convention, the measurements have been carried out under fixed conditions known as standard conditions.

Biochemical standard conditions are usually defined as follows:

- *Temperature: 298 K (25 °C)*
- *Pressure: 1 bar*
- *Solute concentrations: 1 M*
- *pH: 7*

The values of free energy change determined under these standard conditions are expressed as $\Delta G^{o\prime}$ (pH 7 is indicated by the prime symbol).

However, the physiological conditions in which biological processes take place are very different from those established as standard conditions. This can significantly affect the free energy change of a process, even changing its sign and, consequently, its direction. Therefore, it is important to be able to determine the free energy change of a reaction under any given condition different from the standard.

There is an equation that allows the free energy change to be calculated under any set of conditions of substrate concentration and temperature, provided that we know the standard value. For the reaction A + B ⇆ C + D:

$$\Delta G = \Delta G^{o\prime} + R \cdot T \cdot \ln \frac{[C] \cdot [D]}{[A] \cdot [B]}$$

R is the gas constant (0.008314 $kJ \cdot mol^{-1} \cdot K^{-1}$), and T is the absolute temperature in kelvins.

As can be seen in the equation, the free energy change of a reaction will depend on the concentrations of the reaction components. Therefore, as the reaction proceeds, ΔG will change.

For example, the isomerization reaction of fructose 6-phosphate to glucose 6-phosphate releases 1.7 kilojoules for each mole of substrate isomerized under standard conditions:

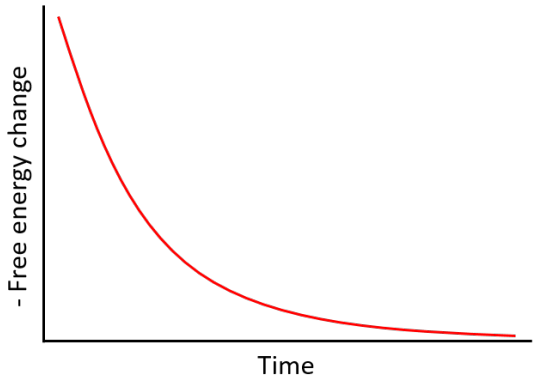
Fructose 6-phosphate ⇆ glucose 6-phosphate $\Delta G^{o\prime}$ = -1.7 kJ/mol

Suppose that the reaction starts with both compounds at a concentration of 1 M. After some fructose 6-phosphate has been converted into glucose 6-phosphate, their concentrations are 0.8 M and 1.2 M, respectively. What is ΔG under these conditions?

$$\Delta G = -1.7 + 0.0083 \cdot 298 \cdot \ln \frac{1.2}{0.8} = -0.7\ kJ/mol$$

The process will continue towards the formation of glucose 6-phosphate until it reaches equilibrium.

We can conclude that In a closed system, **a reaction with ΔG < 0 proceeds in the forward direction until equilibrium is reached** and ΔG becomes 0. In any case, in living cells, many reactions are maintained away from equilibrium by the continuous production and consumption of metabolites.

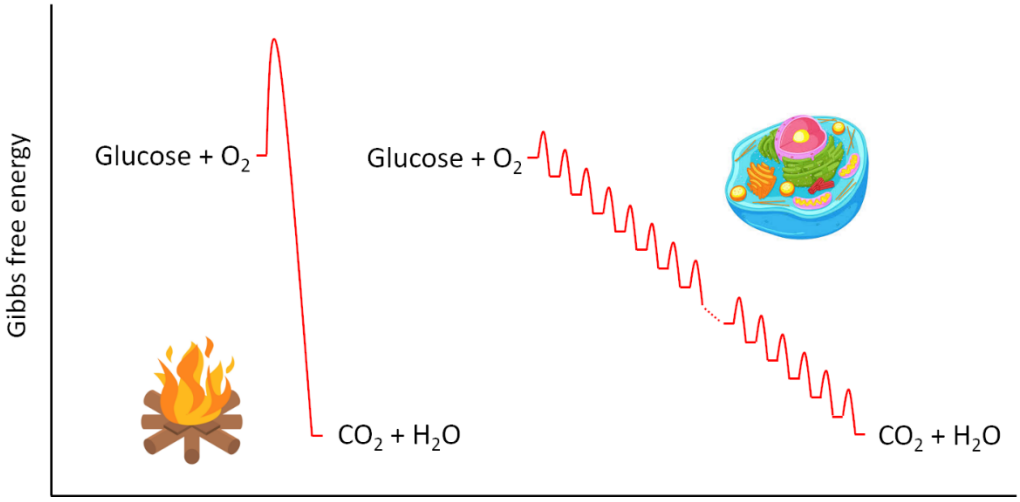


Practical aspects of the concept of free energy change

The most important practical aspect has already been indicated above. The change in free energy shows the direction in which the reaction will proceed spontaneously.

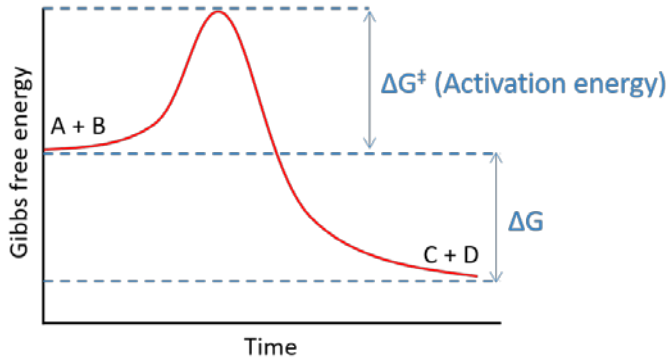
- If ΔG<0, the reaction proceeds in the forward direction
- If ΔG>0, the reaction proceeds in the forward direction
- If ΔG=0, the reaction is in equilibrium

Another important characteristic is that ΔG is independent of the path that the chemical process follows. For example, the free energy that is released when glucose undergoes complete oxidation to carbon dioxide and water (Glucose + $6O_2 \leftrightarrows 6CO_2 + 6H_2O$) is the same when it is burned in a fire as it is when it is enzymatically degraded in a cell.



The overall Gibbs free-energy change associated with the complete oxidation of glucose is independent of the pathway followed, provided that the initial and final states are the same. During combustion, most of the energy is rapidly released as heat. In cells, glucose oxidation occurs through a sequence of enzyme-catalyzed reactions, allowing part of the released free energy to be conserved temporarily in ATP and reduced electron carriers.

The glucose example illustrates another feature of ΔG that should always be considered when analyzing a reaction. **The free energy change provides information about the direction of a reaction, not its rate.**



The rate of a reaction depends on the activation energy. The higher the activation energy, the lower the rate.

Enzymes speed up reactions by lowering activation energies. However, enzymes cannot modify ΔG, so accelerate both the forward and reverse reactions. **Enzymes cannot make a thermodynamically unfavorable reaction favorable**.

Important: All chemical processes that take place in a cell are exergonic

Then, how does a cell manage to make an unfavorable (endergonic) but necessary process take place without contradicting the laws of thermodynamics?

The cell enables an endergonic reaction to proceed by coupling it to a sufficiently exergonic reaction, providing a sufficiently negative free-energy change so that the ΔG of the overall process is negative.

For example, the first step in the catabolism of glucose is its phosphorylation:

$$Glucose + P_i \leftrightarrows glucose\ 6\text{-}phosphate + H_2O$$

Under biochemical standard conditions, this reaction is strongly endergonic ($\Delta G^{o\prime}$ = +13.8 kJ/mol) and is therefore thermodynamically unfavorable.

To make the process exergonic, the cell couples the phosphorylation of glucose with the hydrolysis of ATP, which is a highly exergonic process ($\Delta G^{o\prime}$ = -30.5 kJ/mol):

$$ATP + H_2O \leftrightarrows ADP + P_i$$

Therefore, the global process will be:

$$Glucose + ATP \leftrightarrows glucose\ 6\text{-}phosphate + ADP$$

Since **free energy changes are additive**, $\Delta G^{o\prime}$ of the coupled process will be:

$Glucose + P_i \leftrightarrows glucose\ 6\text{-}phosphate + H_2O$	$\Delta G^{o\prime}$ = +13.8 kJ/mol
$ATP + H_2O \leftrightarrows ADP + P_i$	$\Delta G^{o\prime}$ = -30.5 kJ/mol
Glucose + ATP ⇆ glucose 6-phosphate + ADP	**$\Delta G^{o\prime}$ = -16.7 kJ/mol**

Therefore, **a thermodynamically unfavorable (endergonic) reaction can take place by coupling it with a thermodynamically favorable (exergonic) reaction**.

Finally, remember that cells accelerate processes using specific enzymes. In this case, an enzyme called **hexokinase**.

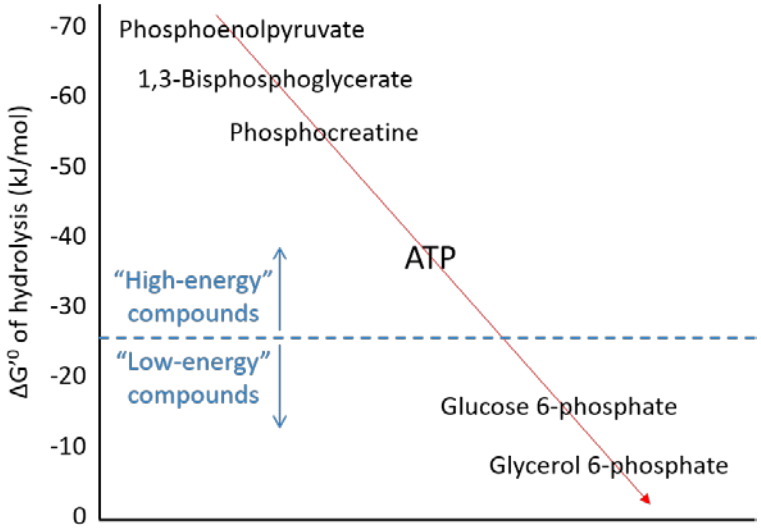
Role of ATP in phosphoryl-group transfer

In the previous example we have seen how ATP acts as a donor of a phosphate group for the synthesis of glucose 6-phosphate. Hexokinase catalyzes the direct transfer of the terminal phosphoryl group of ATP to glucose. Because the phosphoryl group is transferred directly rather than released as free inorganic phosphate, the two reactions are effectively coupled within the enzyme active site. The transfer of phosphoryl groups from ATP is very frequent in our metabolism and is catalyzed by enzymes known as **kinases**.

$$Substrate + ATP \leftrightarrows substrate\text{-}phosphate + ADP$$

The phosphorylated compounds present in living organisms are divided into two groups based on the free energy released after hydrolysis of the phosphate group. Somewhat arbitrarily, those compounds with $\Delta G^{o\prime}$ more negative than -25 kJ/mol are said to be “high-energy” compounds. For “low-energy” compounds, hydrolysis of the phosphate group is less negative than -25 kJ/mol.

For example, ATP is a “high-energy” compound, while glucose 6-phosphate is a “low-energy” compound. “High-energy” compounds are compounds with high phosphoryl-transfer potential. Thus, the hydrolysis of ATP can be coupled to the synthesis of “low-energy” compounds such as glucose 6-phosphate.



On the other hand, there are compounds with greater potential for transferring phosphate groups than ATP. The hydrolysis of these compounds provides enough energy for the formation of ATP. For example:

Phosphocreatine + ADP ⇆ *creatine + ATP* $\Delta G^{o'} = -12.5$ kJ/mol

This reaction is very relevant physiologically since phosphocreatine is stored in skeletal muscle to generate ATP quickly and anaerobically.

There are other reactions throughout metabolism in which "high-energy" compounds are used for the synthesis of ATP. For example, in glycolysis, ATP is synthesized through this reaction:

Phosphoenolpyruvate + ADP ⇆ *pyruvate + ATP* $\Delta G^{o'} = -31.4$ kJ/mol

This strategy for ATP synthesis is known as **substrate-level phosphorylation**.

Finally, **transphosphorylations between nucleotides** are also frequent in metabolism. These reactions make it possible to maintain balanced levels of the different nucleotides.

ATP + NDP ⇆ *ADP + NTP* *Reaction catalyzed by **nucleoside diphosphate kinase***

ATP + AMP ⇆ *2ADP* *Reaction catalyzed by **adenylate kinase***

Question: hydrolysis of ATP in aqueous solution is highly exergonic:

$ATP + H_2O \leftrightarrows ADP + P_i \quad \Delta G^{o'} = -30.5$ kJ/mol

However, the ATP molecule is stable in our cells. What prevents ATP to release a phosphate group into water and turn into ADP?

Oxidation-reduction reactions in biological processes

Electron transfers (oxidation-reduction reactions), in which one molecule loses electrons (it is oxidized) and another gains them (it is reduced), are of crucial importance in metabolism.

In non-photosynthetic organisms, nutrients are the source of electrons for metabolism. Nutrients are oxidized through catabolic processes and electrons pass into so-called **electron carriers** (described below). The carriers, in turn, give the electrons to acceptors with a higher affinity. In aerobic metabolism, the electrons are ultimately transferred to O_2, which acts as the terminal electron acceptor. This flow of electrons provides energy to synthesize ATP.

As with the principles of thermodynamics, the principles of electrochemistry also apply to living organisms.

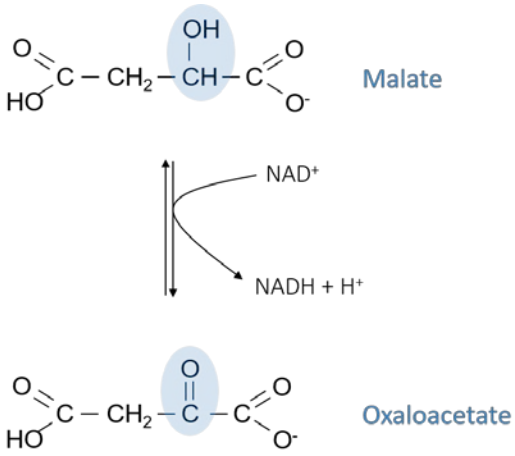
In any oxidation-reduction reaction, two half-reactions (redox pairs) can be defined. For example, the oxidation-reduction reaction $Fe^{2+} + Cu^{2+} \leftrightarrows Fe^{3+} + Cu^{+}$ can be divided into:

$$Fe^{2+} \leftrightarrows Fe^{3+} + e^{-}$$

$$Cu^{2+} + e^{-} \leftrightarrows Cu^{+}$$

In this case, if the reaction proceeds to the right, Fe^{2+} acts as an electron donor and is oxidized. Cu^{2+} acts as an electron acceptor and undergoes reduction.

The oxidation-reduction reactions of organic molecules follow the same logic. For example, in the citric acid cycle, the following reaction occurs, catalyzed by **malate dehydrogenase**:



The C2 alcohol group of malate is oxidized to a carbonyl group. Therefore, malate functions as an electron donor (undergoes oxidation) and NAD^+ as an acceptor (undergoes reduction).

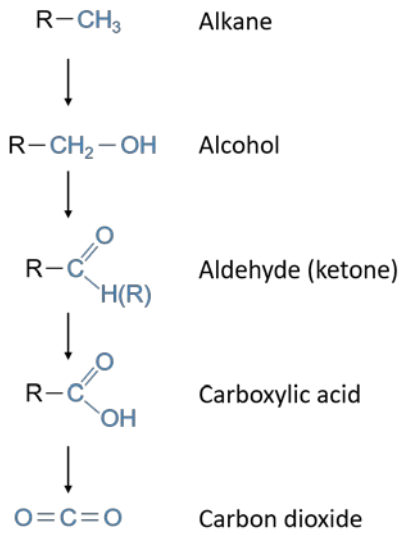
The two half-reactions are as follows:

$$Malate \leftrightarrows oxaloacetate + 2e^{-}$$

$$NAD^{+} + 2e^{-} \leftrightarrows NADH + H^{+}$$

Carbon oxidation states

The carbon atom in biomolecules can be in different oxidation states, ranging from an alkane to carbon dioxide. **It is important to remember the sequence of the oxidation states of carbon in order to be able to understand the oxidation-reduction steps in our metabolism.**



Electron carriers: NADH, NADPH, $FADH_2$ and $FMNH_2$

Hundreds of enzymes in our metabolism catalyze oxidation-reduction reactions in which the electrons from the oxidation of carbon atoms are transferred to **electron carriers**. The reduction of these carriers in catabolic processes allows the conservation of the free energy that is released from the oxidation of substrates. In aerobic metabolism, some of these reduced electron carriers transfer their electrons to the respiratory chain, which ultimately transfers them to O_2. Others, especially NADPH, provide reducing power for biosynthetic processes and other cellular reactions.

NAD^+, **$NADP^+$**, **FMN** and **FAD** are the main electron carriers of our metabolism. These molecules are cofactors of enzymes of the oxidoreductase family, most of which are called **dehydrogenases**. NAD^+ and $NADP^+$ usually act as cosubstrates, diffusible coenzymes that move between enzymes. In contrast, FAD and FMN are generally tightly bound to their enzymes and frequently function as prosthetic groups.

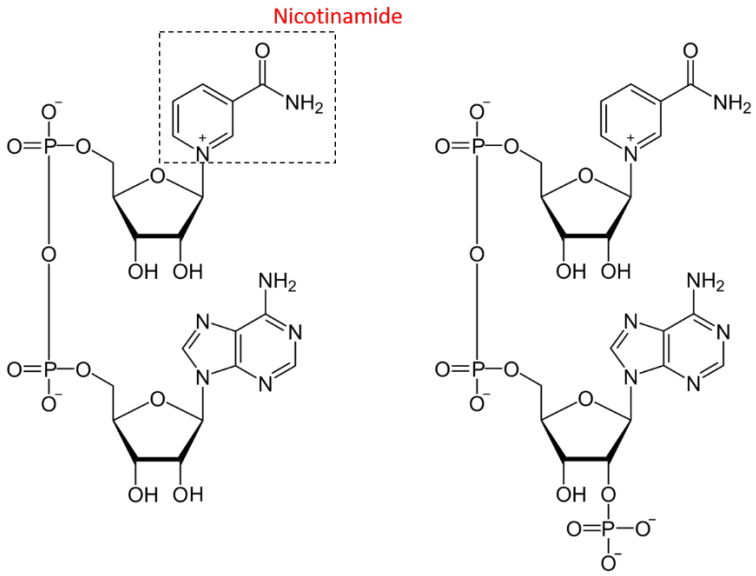
The reactions catalyzed by these enzymes have this structure:

$$AH_2 + \text{oxidized carrier} \leftrightarrows A + \text{reduced carrier}$$

NAD^+ and $NADP^+$

NAD^+ (nicotinamide adenine dinucleotide) and **$NADP^+$** (nicotinamide adenine dinucleotide phosphate) are composed of two nucleotides linked through their phosphate groups by a

phosphoanhydride bond. Because the nicotinamide ring looks like pyridine, these nucleotides are called pyridine nucleotides. **NAD^+ is the main electron acceptor in nutrient oxidation**.



Oxidized forms of the carriers NAD^+/NADH and $NADP^+$/NADPH. The active group is highlighted in NAD^+

The active part of these two carriers is the **nicotinamide** ring. When a substrate molecule is oxidized, it transfers a hydride ion, consisting of one proton and two electrons, to NAD^+ or $NADP^+$, which is reduced to NADH or NADPH, respectively. The second proton is released into the medium.

$$AH_2 \leftrightarrows A + 2H^+ + 2e^-$$

$$NAD(P)^+ + 2H^+ + 2e^- \leftrightarrows NAD(P)H + H^+$$

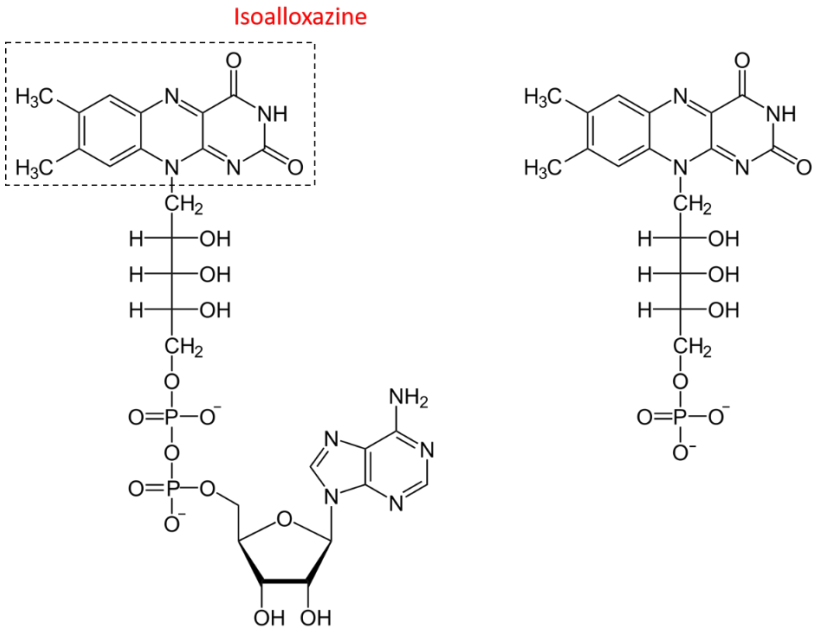
$$\boldsymbol{AH_2 + NAD(P)^+ \leftrightarrows A + NAD(P)H + H^+}$$

NAD^+ and $NADP^+$ are synthesized from **niacin**, a term that includes nicotinic acid and nicotinamide. Niacin can also be synthesized from tryptophan, but endogenous synthesis is generally insufficient to meet physiological requirements, and it is classified as vitamin B_3. Its deficit produces **pellagra**, associated with diarrhea, dermatitis, dementia and, in untreated severe cases, death.

Numerous oxidoreductases use the NAD^+/NADH or NADP+/NADPH pairs as coenzymes. For example, **malate dehydrogenase** and **lactate dehydrogenase** use the NAD^+/NADH pair. **Glucose 6-phosphate dehydrogenase** uses $NADP^+$ as an electron acceptor, whereas **squalene synthase** uses NADPH as an electron donor. In general, **NAD^+ participates in catabolic processes**, whereas **NADPH provides reducing power for anabolic processes** and other cellular functions.

FAD and FMN

Flavoproteins are enzymes that catalyze oxidation-reduction reactions using **FAD** (flavin adenine dinucleotide) or **FMN** (flavin mononucleotide) as coenzyme.



Oxidized forms of the carriers $FAD/FADH_2$ and $FMN/FMNH_2$. The active group is highlighted in FAD.

The group involved in the oxidation-reduction process is the **isoalloxazine** ring, which has the ability to accept and transport two hydrogen atoms, that is, two protons and two electrons, from a substrate that undergoes oxidation.

$$FAD + 2H^+ + 2e^- \leftrightarrows FADH_2$$

FAD and FMN participate in a wide variety of oxidation-reduction reactions. One common example is the formation of a carbon-carbon double bond by dehydrogenation:

$$R–CH_2–CH_2–R' + FAD \leftrightarrows R–CH=CH–R' + FADH_2$$

These two coenzymes are derived from **riboflavin** or vitamin B_2. Riboflavin deficiency, known as **ariboflavinosis**, can cause oral, skin and eye disorders and, in severe cases, anemia.

Some examples of flavoproteins are **succinate dehydrogenase**, **acyl-CoA dehydrogenase**, **mitochondrial glycerol 3-phosphate dehydrogenase** and **NADH dehydrogenase**.

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